

Ranjeet Kumbhar

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Research Interests

My research interests broadly include Applied Artificial Intelligence, Machine Learning, Deep Learning, Exoskeletons and Robotics. I am keen to explore new research directions and integrate cutting-edge techniques across disciplines to advance intelligent systems — and ultimately, build AI that works in the real world.

Education

Master of Engineering in Computer Science & Engineering 2024 – 2026

Thapar Institute of Engineering & Technology, Patiala

CGPA: 9.0

Thesis: *AI-Enabled Adaptive Controller for Lower-Limb Exoskeleton Synchronous Motion in Healthy Elderly Individuals*

- Implemented model-free adaptive control and reinforcement learning for sit-to-stand and walking gait tasks.
- Developed and validated high-fidelity robot dynamic simulations using OpenSim, MATLAB/Simulink.

Bachelor of Engineering in Information Technology (Honours: AI/ML)

2020 – 2024

Savitribai Phule Pune University, Pune

CGPA: 9.36

Publications

R. Kumbhar *et al.* “Sensitivity-Driven Deep Learning Model for Tribological Prediction in Al7075/B₄C Nanocomposites,” *J. Mater. Res. Technol.*, 2026. [DOI]

R. Kumbhar *et al.* “Tribo-Informatics Method for Predictive Modeling of Friction and Wear Mechanisms in Al7075/B₄C Metal Matrix Composites,” *J. Mater. Eng. Perform.*, 2026. [DOI]

R. Kumbhar *et al.* “A Novel Hybrid Controller for Sit-to-Stand Assistance using a Simscape Model based Lower-Limb Exoskeleton,” *Proc. Int. Conf. Modelling, Identification and Control (ICMIC)*, 2026. [DOI]

R. Kumbhar *et al.* “Deep RL-Tuned Model-Free Adaptive Control for Lower-Limb Exoskeletons During Sit-to-Stand Transitions,” (*Manuscript Under Review*), 2026.

Research Experience

Graduate Student Researcher, Thapar Institute of Engineering & Technology, Patiala, India Aug 2025 – July 2026

Research Intern, College of Military Engineering (Indian Army), Pune, India

Dec 2022 – Mar 2023

- Designed a blockchain-based secure file-sharing system deployed in a private distributed LAN environment
- Implemented cryptographic security measures ensuring end-to-end data integrity and access control

Projects

CataractAI – Deep Learning-Based Cataract Prediction System

[GitHub]

- Medical AI system using computer vision on fundus images for cataract detection, achieving **>90% accuracy**
- Implemented EfficientNetB0 with transfer learning (ImageNet pre-training)

Technical Skills

Programming Python, C, C++, SQL, HTML/CSS

ML / DL Deep Learning, Computer Vision, NLP, Statistical Learning, Sensitivity Analysis

Core CS Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS

Frameworks/Libraries MATLAB/Simulink, TensorFlow, PyTorch, Scikit-learn

Achievements & Scholarships

GATE 2024 Data Science and Artificial Intelligence, Score: 422 | All India Rank: 4003

AICTE & TIET Postgraduate Scholarship — INR 3,82,800

SBIF Asha Scholarship Postgraduate Scholarship — INR 3,20,000

Stanford & Coursera Machine Learning Specialization; TensorFlow Developer Specialization

References

Dr. Ashish Singla

Professor, Dept. of Mech. Engg.

Thapar Inst. of Engg. & Tech.

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Dr. Ravinder Kumar

Associate Prof., Dept. of Comp. Sci. & Engg.

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Dr. Rajmeet Singh

Mechatronics Engineer, Research Office

MBZUAI, UAE

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